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Student Number

Year 12, 2023 HSC Mathematics Advanced Assessment Task 1 (38 Marks)

Sequences and Series

Series and Finance

Discrete Probability Distributions

Graphs and Equations

Staff Involved:

8:30 AM

DXC JAI LAK LMD ARM DZB*

PDJ ARP ALY RJW

WEDNESDAY 8th FEBRUARY

155 copies

General

Instructions:

- Working time - 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided separately
- Show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working

1. Circle all tables below that are **NOT** valid discrete probability distributions.

2 marks

x	0.1	0.2	0.3	0.4
$P(X = x)$	0.2	0.3	0.5	0.1

x	1	2	3	4
$P(X = x)$	0.25	0.25	0.25	0.25

x	5	10	15	20
$P(X = x)$	-0.1	0.2	0.4	0.5

x	-9	-3	3	9
$P(X = x)$	0.5	0	0	0.5

2. The overall financial return in dollars on a game of chance is shown in the complete discrete probability table below. Positive x values indicate financial gain, and negative x values indicate financial loss.

x	-4	0	3	9
$P(X = x)$	$4a$	$3a$	$2a$	a

i) Find a as a decimal.

1 mark

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ii) Calculate the expected value of the distribution.
Would you expect to make money playing this game?

2 marks

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- [illegible]

- 1 mark**

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- 2 marks**

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- $$S_n = \frac{n}{2}[2a + (n - 1)d]$$

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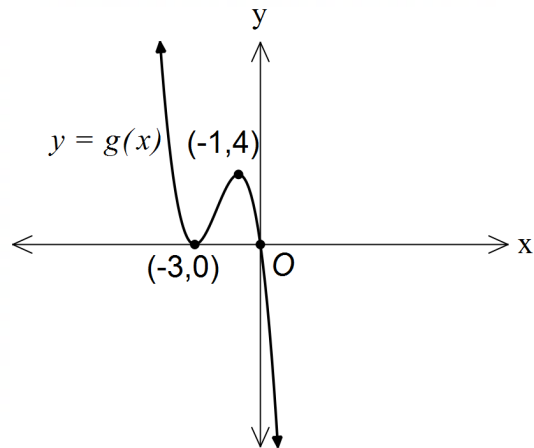
- 2 marks**

[illegible]

- 2 marks**

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- 2 marks**



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- Figure 1 illustrates the growth of a branching structure over five stages (Stage 0 to Stage 4). The structure starts as a single vertical line (Stage 0) and grows by adding branches at a 120-degree angle. The branching pattern is recursive, with each branch at a stage further dividing into two sub-branches at the next stage.

$$\frac{1}{2} \cos 60^\circ = \frac{1}{4}. \text{ (You do **NOT** need to prove this)}$$

- 2 marks**

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- 3 marks**

[illegible]

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- i) Show that the amount owing after 2 months immediately after the instalment is paid is: **2 marks**

$$A_2 = 760\,000 \times 1.0025^2 - 3604.01(1 + 1.0025)$$

- ii) Calculate how many years until the loan is paid off. **3 marks**

- 3 marks**

- 3 marks**

Student number

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iii) The interest for each month, I , can be represented as

$$I = A_n \times r,$$

where A_n is the amount owing after the previous instalment and r is the monthly interest rate.

If Zoe made a monthly instalment where 50% of it erased the monthly interest and 50% of it went towards reducing the principal, calculate the amount still owing immediately after the instalment is paid.

2 marks

End of Paper

12 Adv 2023 Ass1 Solutions

①	x	0.1	0.2	0.3	0.4
	P(x)	0.2	0.3	0.5	0.1

has sum of $P(x) = 1.1$ ∴ not a D.P.O

x	5	10	15	20
P(x)	0.1	0.2	0.4	0.5

is not

$$(P(x) \geq 0).$$

$$\textcircled{2} \text{ (i) } \sum P(x) = 1$$

$$4a + 3a + 2a + a = 10a = 1$$

$$\therefore a = 0.1$$

$$\text{ii) } E(X) = \sum xP(x)$$

$$= (-4) \times 0.4 + 3 \times 0.2 + 9 \times 0.1$$

$$= -0.1$$

∴ No you will not make money

$$\text{iii) } Var(X) = \sum E(X^2) - \mu^2$$

$$= (-4)^2 \times 0.4 + 3^2 \times 0.2 + 9^2 \times 0.1 - (-0.1)^2$$

$$= 16.29$$

$$\therefore \sigma = \sqrt{16.29} = 4.036 \text{ (3dp)}$$

iv)	x	-8	0	6	18
	P(x)	0.4	0.3	0.2	0.1

$$xP(x) = -3.2 + 0 + 1.2 + 1.8 = -0.2$$

∴ worse scenario

$$\textcircled{3} a = 4, d = 5, T_{10} = a + 9d$$

$$= 4 + 45 = 49$$

$$\textcircled{4} ar = 12, ar^4 = 15 \textcircled{2}$$

$$\frac{\textcircled{2}}{\textcircled{1}} \text{ gives } = \frac{15}{12} = r^3$$

$$\therefore r^3 = \frac{5}{4}, r = \frac{1}{2}$$

$$\textcircled{5} L = T_n = a + (n-1)d$$

$$\therefore S_n = \frac{n}{2}(a + L)$$

$$= \frac{n}{2}(a + a + (n-1)d)$$

$$= \frac{n}{2}(2a + (n-1)d)$$

$$\textcircled{6} r = \frac{T_2}{T_1} = \frac{2k}{3} \text{ Limiting sum exists for}$$

$$|r| < 1 \text{ i.e. } \left| \frac{2k}{3} \right| < 1$$

$$\therefore -1 < \frac{2k}{3} < 1$$

$$-3 < 2k < 3$$

$$-\frac{3}{2} < k < \frac{3}{2}$$

$$\textcircled{7} f(x) = (x+2)^2$$

$$f(x-3) - 1 = ((x-3)+2)^2 - 1$$

$$= (x-1)^2 - 1$$

$$\therefore \text{vertex} = (1, -1)$$

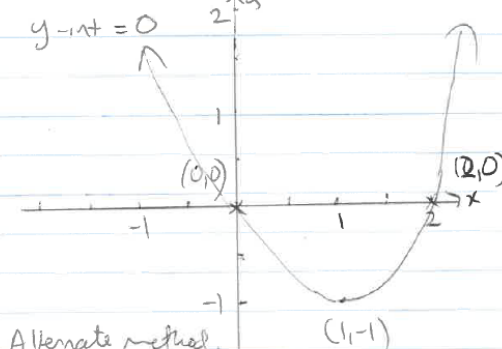
$$x\text{-ints when } y = 0$$

$$\therefore (x-1)^2 - 1 = 0$$

$$(x-1)^2 = 1$$

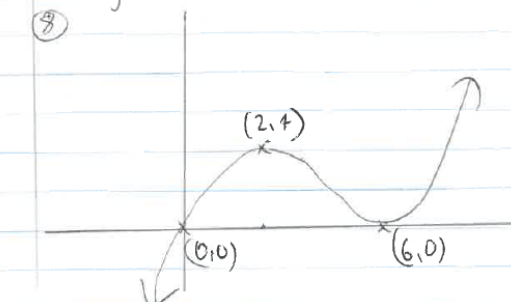
$$x-1 = \pm 1$$

$$x = 1 \pm 1 \text{ i.e. } x = 0 \text{ or } 2.$$



Alternate method.

Sketch $y = (x+2)^2$ then translate 3 right and one down.



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9(c) Vertical height

$$S_1 \quad 1 + \frac{1}{4}$$

$$S_2 \quad 1 + \frac{1}{4} + \frac{1}{4} = 1\frac{1}{2}$$

$$S_3 \quad y = \frac{1}{8} \cos 60^\circ = \frac{1}{16}$$



$$\therefore \text{vertical height} = 1\frac{1}{2} + \frac{1}{16} = 1\frac{9}{16}$$

(ii) Limiting height

$$= 1 + \frac{1}{4} + \frac{1}{4} + \frac{1}{16} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32} + \dots$$

$$= 1 + 2\left(\frac{1}{4} + \frac{1}{16} + \frac{1}{32} + \dots\right)$$

$$S_n, a = \frac{1}{4}, r = \frac{1}{4}$$

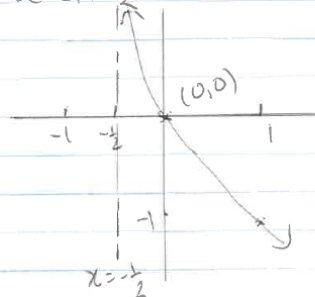
$$\therefore S_n = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$$

$$\therefore \text{limiting height} = 1 + 2\left(\frac{1}{3}\right) = \frac{5}{3}$$

10 Reflect in x-axis

Scale horiz $\times \frac{1}{2}$

Move left $\frac{1}{2}$



$$\textcircled{1} T_5 = a + 4d = 4 \quad \textcircled{1}$$

$$T_2 + T_5 + T_{10} = a + d + 4 + a + 9d = 16$$

$$2a + 10d = 12$$

$$a + 5d = 6 \quad \textcircled{2}$$

$$a + 4d = 4 \quad \textcircled{1}$$

$$\textcircled{2} - \textcircled{1}$$

$$d = 2$$

Continued. Sub $d = 2$ into $\textcircled{1}$

$$T_5 = a + 4d = 4$$

$$a + 8 = 4$$

$$a = -4$$

$$\therefore T_{10} = a + 9d$$

$$= -4 + 18 = 14$$

$$\textcircled{12} A_1 = 760000(1.0025) - 3604.01 \quad (r = \frac{0.025}{12} = 0.0025)$$

$$A_2 = A_1 \times 1.0025 - 3604.01$$

$$= (760000(1.0025) - 3604.01) \times 1.0025$$

$$- 3604.01$$

$$= 760000(1.0025)^2 - 3604.01(1.0025)$$

$$- 3604.01$$

$$= 760000(1.0025)^2$$

$$- 3604.01(1 + 1.0025)$$

$$\textcircled{ii} A_n = 760000(1.0025)^n$$

$$- 3604.01(1 + 1.0025 + \dots + 1.0025^{n-1})$$

$$n = n, r = 1.0025, a = 1 \text{ in } S_n$$

and $A_n = 0$ when loan is paid off.

$$\therefore 0 = 760000(1.0025)^n - 3604.01 \left(\frac{1.0025^n - 1}{0.0025} \right)$$

$$\therefore 1441604 \times 1.0025^n - 1441604$$

$$= 760000(1.0025)^n$$

$$\therefore 681604 \times 1.0025^n = 1441604$$

$$1.0025^n = \frac{1441604}{681604} = 2.1166$$

$$\therefore \ln 1.0025 = n = 299.99 \dots$$

$$\ln \left(\frac{1441604}{681604} \right) = 299.99 \dots \therefore 300 \text{ months} = 25 \text{ years}$$

$$\textcircled{iii} I = A_n \times r, r = 0.0025, I = \frac{1}{2}(3604.01)$$

$$\therefore 1802.005 = A_n \times 0.0025$$

$$\therefore A_n = \$720802 \text{ is amount owing}$$

after previous instalment.

$\therefore$ Amount owing after this instalment is

$$A + I - M = 720802 + 1802.005 - 3604.01$$

$$= 718999.99$$